1 Introduction to CCD's

1.1 A brief history of astronomical detectors

1.1.1 The Human Eye

Throughout most of history, only one type of detector has been available to us for astronomical observation: the *human eye*. The hunan eye is a very versatile detector, but —despite its strengths— has several limitations as well. Since in modern astronomy we still encounter some concepts that are intimately tied to the earlier use of this type of detector (most notably the continued use of *magnitude*s), let us discuss some of the properties, pros and cons, of the human eye. A schematic drawing of the human eye is shown in Fig. 1.

- The amount of light that can be collected by a detector depends on the aperture through which light enters. For a well-accommodated eye, the entrance aperture (i.e., the *pupil*) diameter can be as large as ∼8 mm (∼$\frac{1}{3}$ inch). This limitation can be overcome, however, by using a telescope with a larger entrance aperture to focus more light through the pupil.

- Although the most sensitive cells within the retina, the *rod cells*, have a finite probablity of responding to the absorption of even a single photon, the overall probability of absorption and response is only ∼1–2%.

▷ The ratio of incoming to actually detected photons is denoted by the term *quantum efficiency (QE)*, and is usually expressed as a percentage.

- Another limitation is that the human eye has a fixed response time (taken here to be equivalent with an *integration time*) of roughly $\frac{1}{10}$ second (rod cells), followed by a recuperation period (*dead time*) of similar length, during which a given cell cannot respond to further incoming photons.

- The eye plus brain is a *logarithmic* detector: its response is not proportional to the incoming photon flux, but to the logarithm thereof.

- The range in wavelengths to which the human eye is sensitive is small. Rod cells, responsible for night vision, are only sensitive to wavelengths shorter than ∼570 nm. The short wavelength cut-off depends on age, but is generally longward of ∼370 nm (see Fig. 2).

▷ The wavelength range over which the quantum efficiency of a detector exceeds some fiducial fraction of the peak QE value is denoted by the term *band-pass*. The fiducial fraction is commonly taken to be 10%.

© 2006 R.A. Jansen

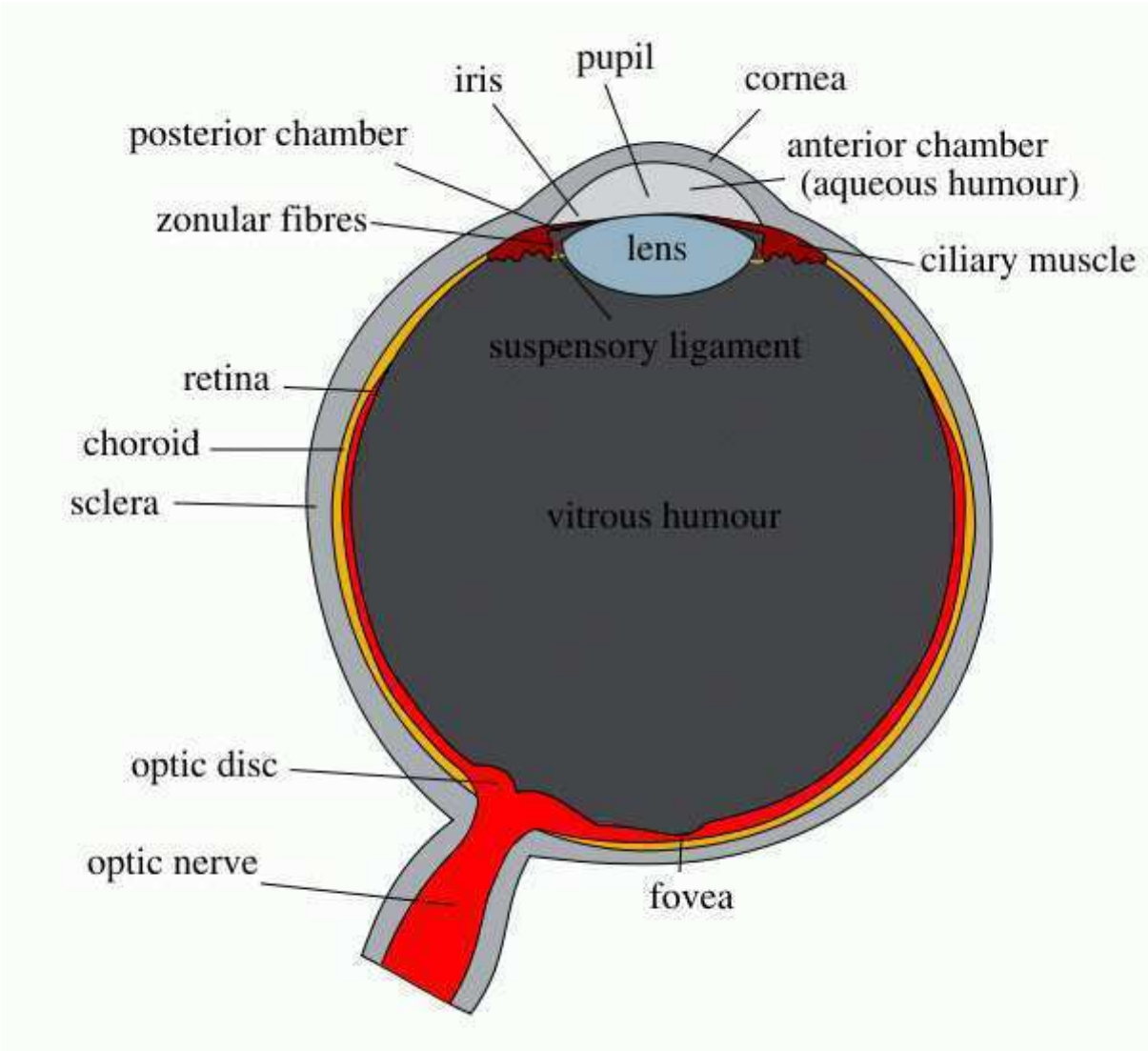


Figure 1: (a) Schematic diagram of the human eye. In a well-accomodated human eye, the diameter of the pupil (opening aperture) can reach ∼8 mm. The optic disk is better known as the *blind spot*, where the optic nerve enters the eye. Averted vision will optimize our use of the more sensitive rod cells, which are located in the retina mostly outside the fovea. The less sensitive cone cells, responsible for our perception of color, are strongly concentrated toward and densely packed within the fovea. This dense packing of photoreceptive cells also gives us our visual acuity (sharpness of vision). *Courtesy: Wikipedia.*

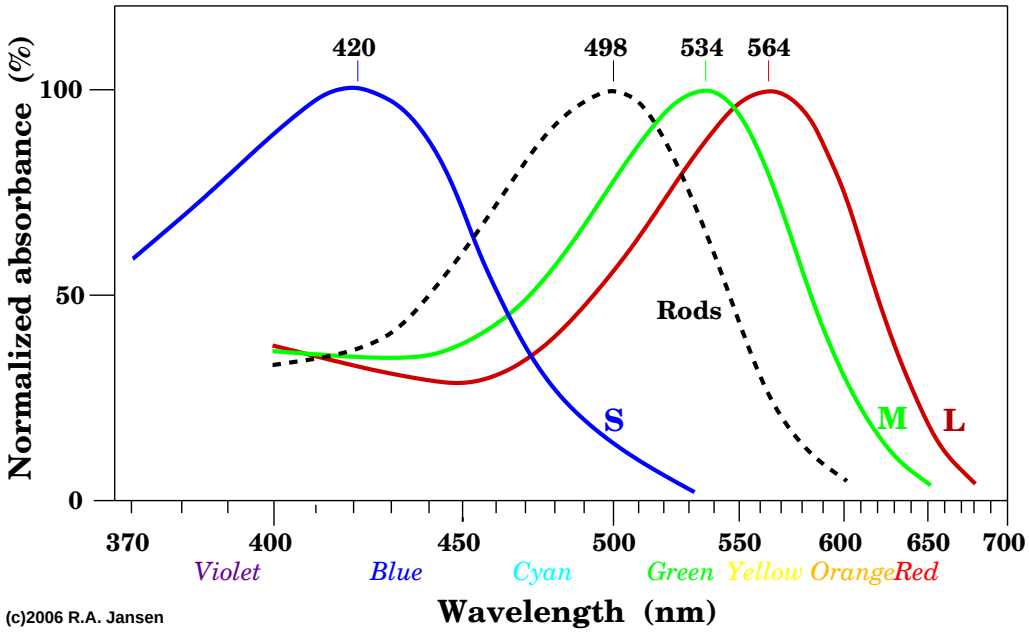


Figure 2: Normalized spectral absorption curves of the pigment in human rod cells (responsible for night vision) and of the short (S), medium (M) and long (L) wavelength pigments in cone cells. The sensitivity of the rod cells is largest in the blue-green portion of the optical spectrum. The absolute sensitivity of the rod cells is ∼100 times higher than that of the cone cells. *After Bowmaker & Dartnall 1980.*

- Thanks to the two different types of photo-receptive cells and their "inverse" physiology (an *increase* in the light level results in a *decrease* of the amount of neurotransmitters released, and high light levels cause the photo-destruction of the absorbing pigment), and thanks to an adjustable entrance aperture, the eye can function well over a very large range in brightness. Simultaneous detection of both bright and faint light sources can be a problem (e.g., think of the loss of night vision experienced due to head-lights of oncoming cars at night: it takes time for the cell to produce new absorbing pigment).

Magnitudes

The ancient Greeks (notably Hipparchus) made an inventory of stars in the sky, where the 20 brightest stars were defined to be of magnitude 1. Fainter stars were divided into 5 bins according to their perceived visual intensity, with roughly equal steps in perceived brightness between the first and subsequent bins. In more recent times, in 1856, Pogson derived his famous quantitative expression to approximate the classical magnitude scale, under the assumption that the human eye behaves as a logarithmic detector:

$$M = -2.5 \cdot \log F + C \quad ,$$

where F is the flux arriving from a star and C is an arbitrary zeropoint. The scale factor -2.5 is called the *Pogson scale*. The magnitude *difference* between two stars is then related to the *ratio* of their fluxes by:

$$M_1 - M_2 = -2.5 \cdot \log(F_1/F_2).$$

Note, that a flux ratio of 100 corresponds exactly to a difference of 5 magnitudes. To this day, astronomy students drive physics students nuts by putting seemingly arbitrary factors of -2.5 before any decimal logarithm they encounter.

© 2006 R.A. Jansen

1.1.2 The Photographic Plate

During the second half of the 19th century, the then new technology of photographic emulsions found its first astronomical uses. The earliest known photograph of an astronomical object, the Moon, was made by J.W. Draper in 1840, followed five years later by Foucault and Fizeau, who photographed the Sun. By 1870, technology had improved enough to allow Jansen & Lockyer to discover the element Helium on a photograph of a solar spectrum.

With the advent of photography, it became possible to overcome the fixed exposure time of the human eye, and to faithfully record detected photons for later analysis. It also became possible to record photons at wavelengths for which the human eye is insensitive, particularly toward shorter wavelengths (near-UV; but also X-rays and ionizing particles). Photographic emulsions sensitive to longer wavelengths (near-IR) were only developed later in the 20th century.

The glass plates used as substrates for the emulsions were resistent to bending and distortion during the observations, and have proven *extremely* stable over time (a full century and counting). Although currently available in digitized format, the two Palomar Observatory Sky Surveys (POSS in the 1950s, and POSS-II during the 1990s) and the complementary UK Schmidt surveys in the southern hemisphere were made using photographic plates.

- The quantum efficiency of normal photographic plates is low, typically only 1–2% ($\lesssim$3% for IIIaJ). Via a black magic process of *hypersensitization*, involving exposure to different gasses at various temperatures, this could be increased for some purposes to $\sim$10%, but at the cost of a reduced *dynamic range* (even in the absence of any photons, such plates may show a distinct "fog").

▷ The ratio of the maximum to the minimum measurable signal is denoted by the term *dynamic range*.

- Photographic plates are non-linear/logarithmic in their response to light. Over a limited range, between under- and overexposure, the response of photographic plates is close to logarithmic (see Fig 3). Outside this range, the response is highly non-linear. The effective dynamic range is typically only $\sim 10^2$.

  Given the very large range in brightness of astronomical sources that might be of interest within a given field of view on the sky, and even within individual extended objects such as galaxies and nebulae, often a sequence of different exposure times is required. For long exposures, *reciprocity failure*, the decrease of the sensitivity as the exposure continues, could be an issue.

- Quantitative analysis of photographic plates requires one to first scan and digitize those plates. This digitization process is far from trivial and not always perfectly repeatable.

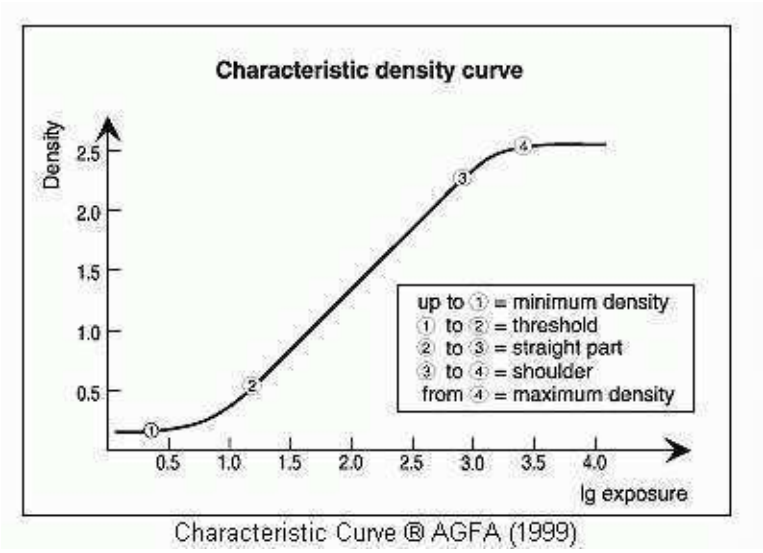


Figure 3: Schematic representation of a characteristic density curve of photographic emulsions. Between points (2) and (3), the response to light is very close to logarithmic.

- Photographic plates were the first detectors to record detected photons for later detailed study and their long-term stability is excellent. We can still compare present-day observations with plate material from 1900.
- Photographic plates have an extremely large area — unmatched to this day by any other type of 2-D detector, and well-matched to the physical size of the collimated beam of a large telescope ($\gtrsim$10 inch). In combination with Schmidt telescopes, typical areal coverage was $6^\circ \times 6^\circ$!
- The resolution of a photographic plate, defined by the grain size of the silver halide crystals (a few μm), is very high. A photographic plate has a higher resolution than a CCD chip of the same area.

The sensitivity of early photographic plates to blue light results to this day in the preponderance of blue photometric and spectroscopic standard stars.

Silverhalide photography

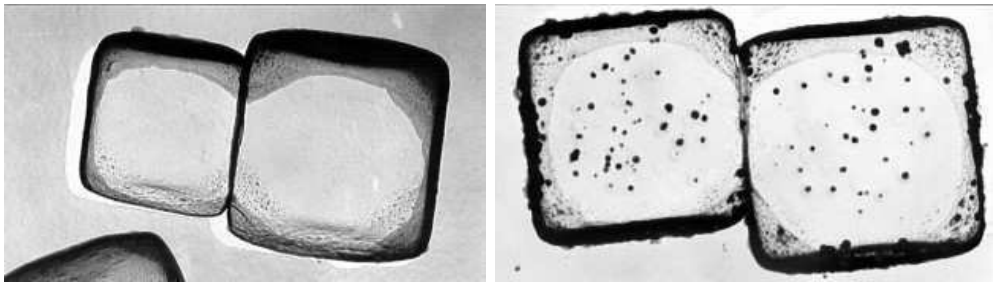


Figure 4: (a) unexposed silverbromide crystals, (b) exposed silverbromide crystals, where the metallic silver deposits are clearly seen (even after only soft development). *Courtesy: Frank Lakiere.*

Photochemical reaction equation for silver bromide:

$$\mathrm{Ag^+Br^-(crystal)} + h\nu\,\mathrm{(light)} \longrightarrow \mathrm{Br} + e^- + \mathrm{Ag^+} \longrightarrow \mathrm{Br} + \mathrm{Ag^0}$$

The free silver atom produced within the silver halide grain serves as a catalyst during subsequent development, where grains containing free silver atom(s) will form relatively large deposits of free silver (opaque), while those that do not contain free silver (grains that did not absorb photons) will remain clear.

1.1.3 The Photomultiplier Tube

A major advance for low light-level applications was the development in the first half of the 20th century of vacuum-tube technology, including Photomultiplier tubes (PMTs). These detectors rely on the *photo-electric effect* and can achieve peak quantum efficiencies of upward of 20%. They are usable over a very wide range in wavelength. This type of detector *amplifies* the exceedingly faint signal of a single absorbed photon (i.e., the production of a photo-electron) by causing a cascade of secondary electrons. Amplification can be as high as a factor 10^8 (typically $\gtrsim 10^6$).

A PMT works as follows (see Fig. 5). When an incident photon strikes the thin photocathode, made of a semiconducting material such as GaAs (galliumarsenide) deposited on the inside of the entrance window, a photo-electron is dislodged. That photo-electron has some residual kinetic energy allowing it to escape from the cathode. A strong electric field (high voltage) accelerates this electron towards the first of a series of electrodes, called *dynode*s, where it arrives with sufficient energy to release a second electron. Both electrons are then accelerated toward a second dynode, each again releasing additional electrons. Each subsequent dynode in the stack is held at a more positive voltage than the previous one. The geometry of the dynode chain is such that a cascade occurs with an ever-increasing number of electrons being produced at each stage. Finally, at the *anode*, the accumulation of charge results in a sharp current pulse, the height of which is proportional to the number of original photo-electrons (and therefore photons).

- Their peak quantum efficiency is much higher than that of photographic plates and can reach $\sim$40%.
- They can be operated as virtually noise-free, highly linear and stable photon counters, allowing accurate photometry and spectrophotometry to accuracies of $\sim$1%, and hence quantitative astrophysics. They also allow high-speed photometry (few μsec).
- Although their *dynamic range* is limited by the build up of charge near the anode, which repulses arriving electrons, their dynamic range tends to be very large: typically $\sim 10^5$.
- Their opening aperture is physically large (from several cm^2 to over 50 cm diameter [the Cerenkov radiation detectors within Super-Kamiokande]) allowing industrial and medical applications (e.g., counting blood cells flowing past at high speed).
- To reduce their *dark current* due to thermal electrons being drawn away from the cathode and dynodes, they must be cooled. Typically, they are operated at dry ice (solid CO_2) temperatures ($-78\,^{\circ}$C).

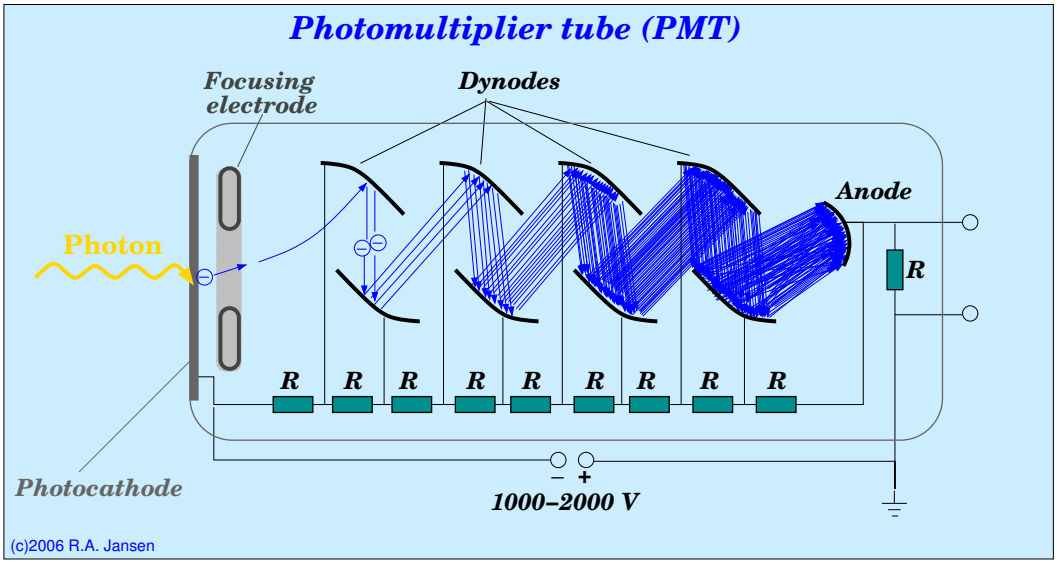


Figure 5: Schematic diagram of a photomultiplier tube. When a photon strikes the thin photocathode on the inside of the entrance window of the glass vacuum tube, a photo-electron is released and accelerated toward the first of a series of dynodes, where it arrives with sufficient energy to dislodge a second electron. The geometry of the dynode chain is such that a cascade occurs with an ever-increasing number of electrons being produced at each stage. The height of the sharp current pulse measured at the *anode* is directly proportional to the number of original photo-electrons (and therefore photons).

- Photomultiplier tubes are high-voltage devices, requiring typically 1000–2000 V for their operation. When in operation, photomultiplier tubes are prone to irreversible damage when exposed to high light levels. They also suffer from electrons dislodged by cosmic rays. Unless shielded, they are sensitive to strong ambient electric and magnetic fields.

- The main disadvantage of photomultiplier tubes, however, is that they are single-element devices and next to impossible to package into useful 1-D and 2-D arrays. They are, therefore, not very efficient for spectroscopy and not useful for astronomical imaging applications (other than in scanning and digitization of photographic plates).

To this day, astronomers and filter engineers are striving to reproduce the response curves, and in particularly the red cut-offs, of a few popular models of photomultiplier tubes, since the sensitivity of these tubes rather than the pieces of colored glass employed defined the effective "filter transmission" curves of the standard Johnson UBV and Kron-Cousins RI filters. (E.g., in the case of Johnson I, the piece of

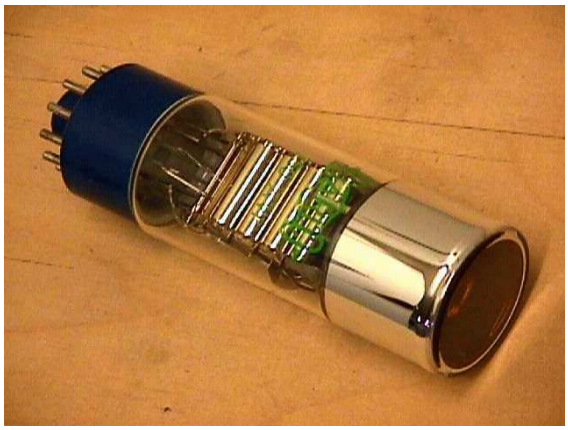


Figure 6: Example of a photomultiplier tube (PMT). *Courtesy: M. Fletcher.*

glass Johnson used was a long-pass filter, defining only the blue portion of the I transmission curve; the red portion reflected the sensitivity of his photomultiplier).

1.1.4 Image Intensified Photography

An attempt to combine the advantages of a photocathode device (photo-electric effect and signal amplification) and the 2-D imaging capability of photographic plates. Although this technology experienced a brief revival with the first baloon-borne and space-based UV imaging telescopes, it should be considered a station that was short-lived — the less said, the better.

1.1.5 Image Intensifier Vacuum Tube technology

A series of high-voltage 1-D and 2-D electronic devices were constructed starting in the 1960s based on image intensified vacuum tubes. Like in the photomultiplier tube, incident photons release photo-electrons which are accelerated and produce either a detectable current pulse or a splash of light when hitting a screen coated with a phosphor (e.g., ZnS:Cu in old-fashioned green computer screens). Unlike photomultiplier tubes, information on where exactly the incident photon arrived is preserved.
Examples: Electron Conduction Vidicon, Intensified Dissector Scanner, Intensified Reticon Scanner, Image Photon Counting System, and Multi-Anode Microchannel Array detector. The latter has experienced a revival in space-based UV astronomy (e.g., HST/STIS and HST/ACS-SBC, GALEX, and FUSE).

- Disadvantages are the high voltages required (e.g, 25 kV), image blur, and non-trivial relation between arrival location and (x, y) position.

1.1.6 The Charge Coupled Device

Charge Coupled Devices (CCDs), were originally developed at Bell labs as a solid-state memory device (Amelio, Tompsett & Smith 1970; Boyle & Smith 1970). A CCD is essentially a regular array of large numbers of coupled Metal-Oxide Semiconductor Field Effect Transistors (MOSFETs). It was immediately realised, however, that the sensitivity to light via the photo-electric effect and the ability to store charge in a regular array format was suitable for application as a 2-D detector. An excellent review paper on the workings of a CCD is MacKay (1986).

When a photon is absorbed in silicon, an electron-hole pair is created. In a CCD, two or more electrodes (called *gates* following MOSFET terminology) per isolated picture element (*pixel*) create potential wells in the silicon. The generated photo-electrons migrate to and are confined within the potential well created by the gate kept at the largest positive Voltage. The holes, on the other hand diffuse away into the bulk of the silicon, thus preventing recombination.

In a three-phase CCD (presently still the most common type in astronomical applications), three gates — each isolated from the silicon substrate and from one another — define a pixel row. Embedded channel stops of heavilly doped p-type Si define pixel columns, and prevent electrons from migrating along the length of the gates due to their negative charge. Fig 7 illustrates the structure of a three-phase CCD, and Fig 8 shows a top view of a portion of an actual device. Every third gate is electrically connected in parallel and kept at the same voltage.

To convert the 2-D array of stored photo-electrons into an image, the charge in each pixel needs to be measured (or more accurately: estimated). To this end the voltages on the electrodes of the array are cycled in such a way that the stored charges are sheparded along the columns, effectively being passed along from one pixel to the next. At the end of a column the charges are passed onto a special set of pixels with a gate structure that is orthogonal to that in the image area of the CCD. Here the charges collected in an entire pixel row are shifted one pixel at a time and passed on to on-chip electronics that amplify and estimate the stored charge. The result is subsequently converted by an Analog-to-Digital circuit into a digital number (refered to as either *Data Number* (DN), or *Analog-to-Digital Unit* (ADU), or simply *Count*).

The most common analogy for the process of reading out a CCD is that of a "Bucket Brigade", where the contents of the buckets on many parallel belts are emptied in a serial belt with one bucket per parallel belt. The buckets on the serial belt are then emptied one at a time into a measuring container. After the last bucket on the serial belt has been measured, the next row of buckets on the parallel belts is emptied into the buckets on the serial belt, and so on (see Fig. 9). A more detailed illustration of how the charges are passed along from pixel to pixel by cycling the gate voltages is given in Fig. 10.

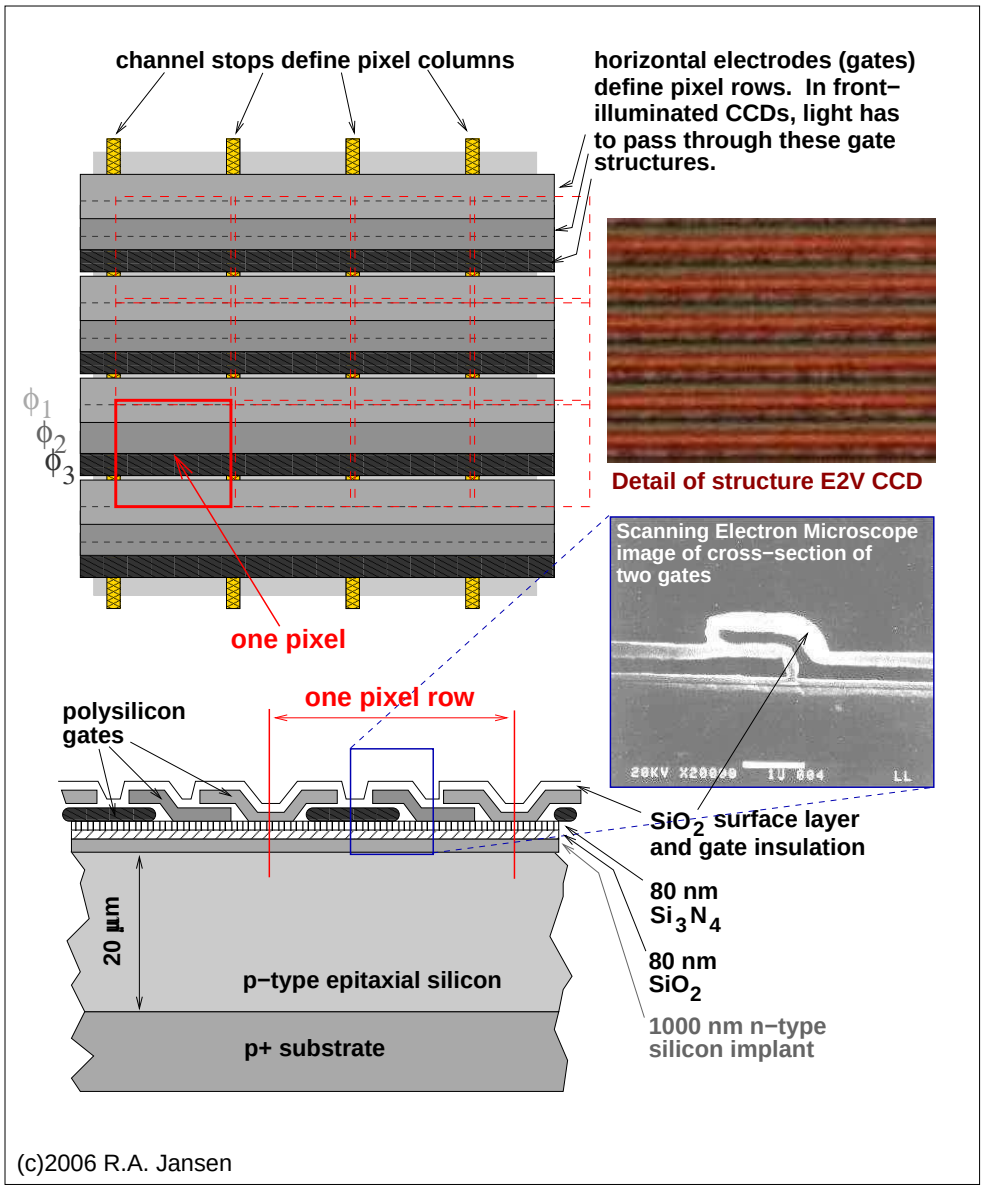


Figure 7: Schematic representation of the gate structures on a typical 3-phase CCD. Each pixel is defined by three gates and two channel stops (buried beneath and electrically isolated from the gates). When the CCD is exposed, all photo-electrons generated within the pixel volume migrate to and remain confined in the potential well of the middle gate, with the largest positive voltage (ϕ_2). The polysilicon gates partially overlap, but are isolated from one another by a thin layer of SiO_2. The relatively thick ($\sim 1\mu$m) layer of n-type Si, below the insulating SiO_2 and Si_3N_4 layers, helps reduce charge transfer losses: it keeps accumulated electrons away from the surface, where some might become trapped. At right, a detail from the image area of the CCD of Fig. 8 is shown. The repeating horizontal pattern of the gates is clear. An electron microscope scan reveals the vertical structure at the overlap of two gates.

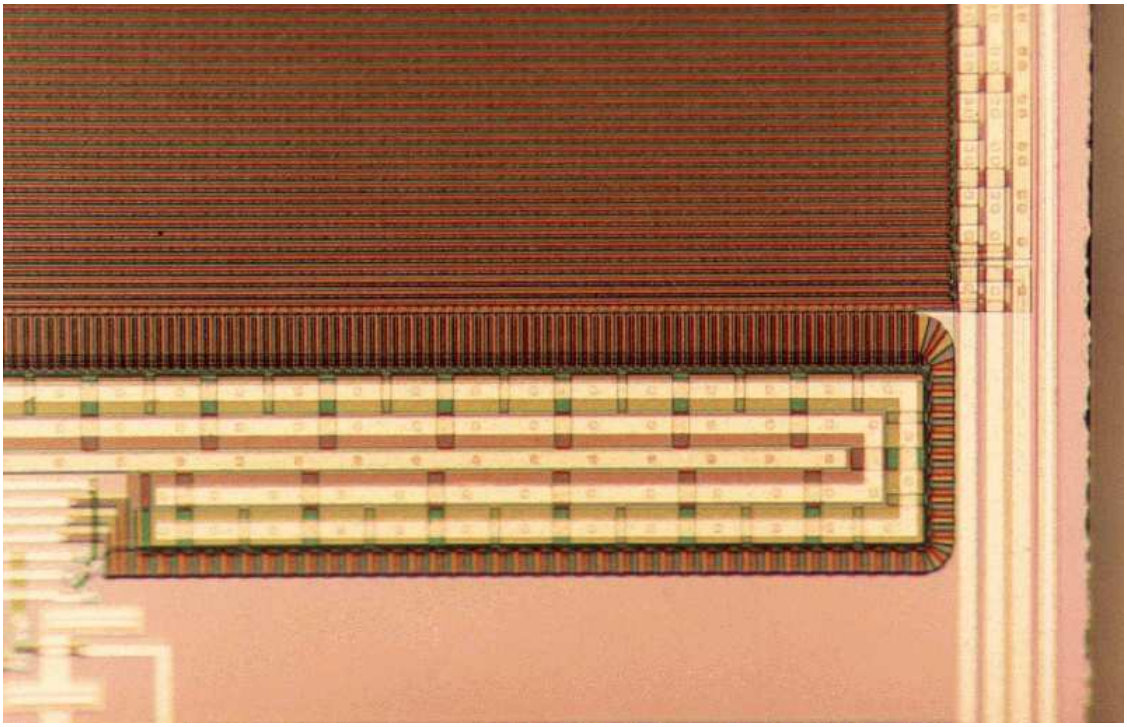
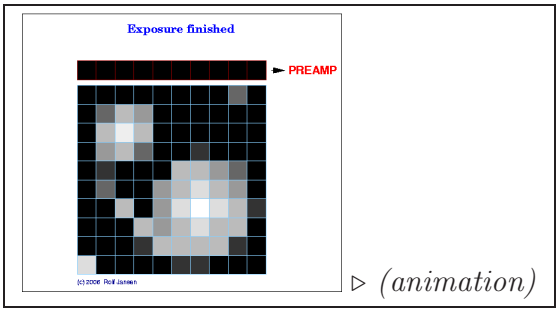


Figure 8: Photomicrograph of a corner of an E2V L3CCD, showing the image area (pixels), and the serial register (the vertically extended pixels just below the image area. On this particular CCD, the serial register bends around into an extended register, where the charge is amplified (clock voltages in the extended portion are sufficiently large to ensure a finite probability of dislodging additional electrons in each charge transfer from pixel to pixel. Part of the actual on-chip read-out electronics is visible at the lower left.

In class, an animation of the operation of a CCD, consisting of preparing the CCD for an exposure (clearing any residual accumulated charge), exposing, and read-out will be show.



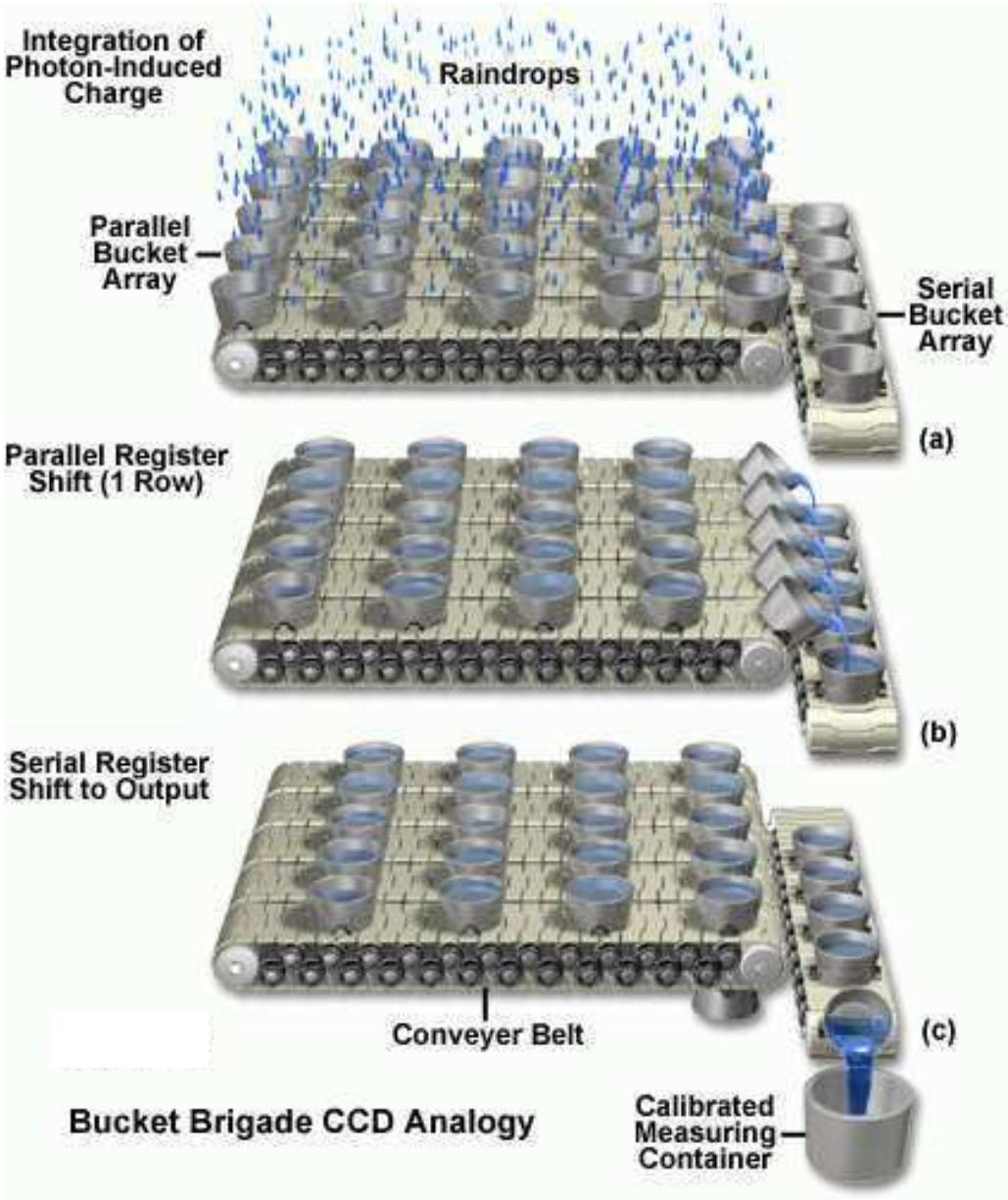


Figure 9: The "bucket brigade" analogy often used to visualize the operation of a CCD. Each bucket represents a pixel that can store charge during integration (exposure of the CCD to light). At the end of the integration, all parallel belts will simultaneously rotate their buckets toward the serial belt and dump the contents of one bucket per belt into the corresponding bucket on the serial belt. Then the serial belt rotates and presents one bucket at a time for measuring. When all buckets on the serial belt have been measured, the parallel belts will rotate and dump the next "row" of buckets into the corresponding buckets on the serial belt, and so on until all buckets have been measured. *Courtesy: Nikon/K.R. Spring, T.J. Fellers & M.W. Davidson.*

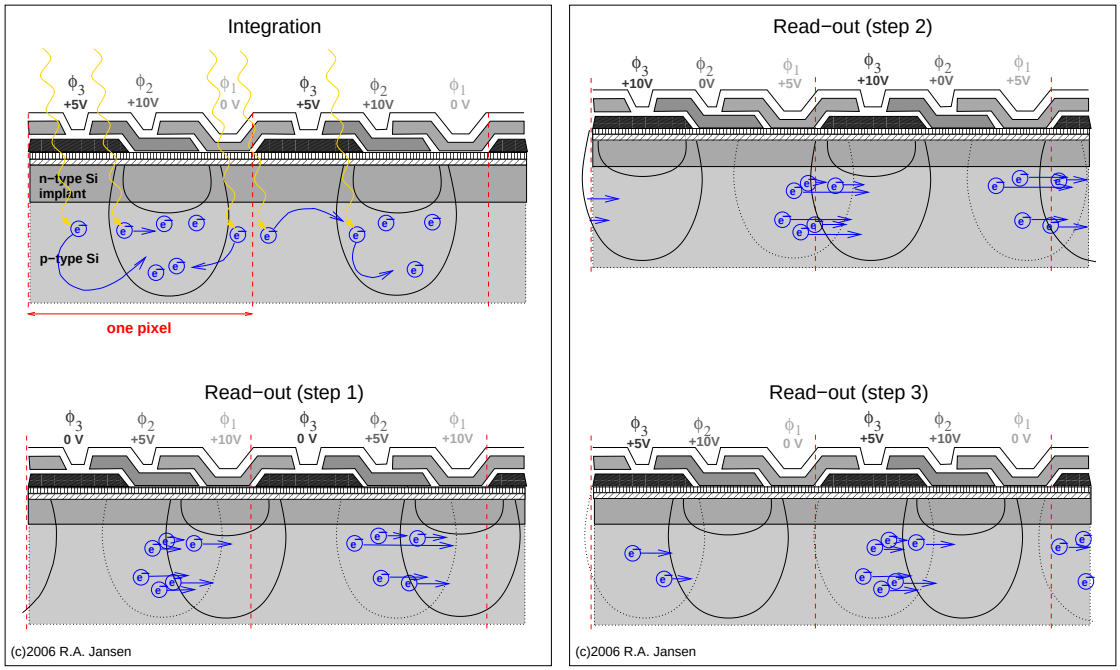


Figure 10: Schematic representation of the operation of a CCD during integration and read-out. During integration, the middle gate (ϕ_2) is kept at a large positive voltage, causing all photo-electrons generated under any of the three gates defining a pixel to migrate to the deep potential well of this gate. Note that the effect of the portion of the middle gate that overlaps adjacent gate ϕ_3 is shielded. When the CCD is read out, the voltages on the gates are cycled in succession, such that the accumulated electrons follow the deepest potential well until, at the end of a single parallel clocking cycle, they have been transferred to the adjacent pixel along the column. The illustration follows the transfer of charge within a single pixel column. Since the voltages on the gates in each pixel row along a column are connected in parallel, a single parallel clocking cycle will transfer the accumulated charge in the pixels of a row to the corresponding pixels of the serial output register (not shown). The charge is then transferred out of the serial register, one pixel at a time, to be amplified and sampled by the A/D converter. This is repeated until the entire array has been read out.

Summary of advantages and disadvantages of CCDs:

- CCDs can have peak QE in excess of 90% (when properly AR coated). Even uncoated, front-illuminated CCDs can have peak QE in excess of 70%. The QE is only a relatively slow function of wavelength within this bandpass.

- Their response to incident photons is highly linear over almost its entire operational regime.

- The natural operational band-pass of CCD's is ∼380 nm–1.1 μm, which can be extended down to ∼300 nm by optimizing antireflection coatings and by etching away most of the silicon substrate (*thinning*) and illuminating the CCD from the back. The band-pass can be extended even further toward the UV using phosphorescent coatings (like the one in your yellow marker pens), and somewhat toward the near-IR by additional doping of the silicon.

- The *full-well capacity*, the maximum number of photo-electrons that can be stored within a given pixel, is typically of the order of 10^5 e^-. The dynamic range is typically $\sim 10^4$. When more charge accumulates than can be confined by the gate potential (*full-well saturation*), the excess charge may bleed into adjacent pixels along a column.

- Saturation may also occur in another way: when the conversion of the measured charge to a digital number would require a larger number than can be stored in the available number of bits (*A/D saturation*).

- CCDs require cooling to cryogenic temperatures using liquid N_2 (−100°C–−120 °C or 170–150 K), or at least thermo-electric cooling (at ∼ −80 °C, the detectors aboard the Hubble Space Telescope are at the warm end of the operational range). This cooling is required to reduce *dark current* due to thermal electrons from the bulk silicon also being dislodged and collected, and to reduce thermal noise in the read-out electronics (both on-chip and in the adjacent "head electronics").

- CCD's have small physical sizes ($\lesssim$4.5 inch) that are poorly matched to the physical size of the collimated beam of large telescopes ($\gtrsim$10 inch diameter), so that only a small portion (solid angle) of the focal plane of the telescope is actually used. CCD sizes are set by the difficulty of producing flawless large-format devices, and by the ∼12 cm diameter of the wafers used for all current silicon-based integrated circuit fabrication. Monolithic devices up to 8192×8192 pixels can now be semi-routinely produced, but the work-horse devices are currently only 2048×4096 pixels. Pixels are typically between 10μm and 24μm in size. The physical size limitation has been partly overcome by constructing arrays of multiple CCDs, that are read out simultaneously. Examples of such arrays are shown in Fig. 11.

- The temporal resolution, set by the maximum rate at which the array can be read out, is $\gtrsim$10 msec, i.e., $\sim$1,000–10,000$\times$ slower than for a PMT. In most integrating applications, the temporal resolution is *much* lower. The read-out time is largely set by the minimum sampling time per pixel to accurately estimate the accumulated charge. (Trade-off: faster read-out $\rightarrow$ less accurate; slower read-out $\rightarrow$ more accurate). Effective read-out times have been reduced in recent years by CCD designs that allow simultaneous read-out of four quadrants of the CCD through serial registers and amplifiers in each corner of the CCD, and through the development of faster and lower-noise read-out amplifiers.

- The *Charge Transfer Efficiency* (CTE) of CCDs is very close to, but less than 1. The charge losses accumulate the further a given pixel is removed from the read-out amplifier. For a 2048$\times$2048 pixel CCD read out through a single amplifier, the charge in the last pixel read experienced 2048 parallel transfers, followed by 2048 serial transfers. If we assume that parallel and serial transfers are equally efficient (not necessarily so), this results in only $(\mathrm{CTE})^{4096} \times 100\%$ of the originally generated photo-electrons to arrive at the read-out amplifier. In modern CCDs with buried channels, CTE is typically $\gtrsim$0.99998, giving $\gtrsim$92% in the above example.
  The complement of CTE, *Charge Transfer Inefficiency* ($\mathrm{CTI} = 1 - \mathrm{CTE}$) quantifies the charge losses. Hence $\mathrm{CTI} \lesssim 8\%$ in the above example. If the same CCD was read through four amplifiers (effectively forming 4 arrays of 1024$\times$1024 pixels), the losses would be $1 - (0.99998)^{2048} \lesssim 4\%$

CCDs in different flavors

CCDs come in several flavors, each optimized for specific uses, and with possible additional electronic circuitry built onto every pixel, or onto the serial output register. Examples are:

- Surface Channel CCDs. The first and simplest type of CCD, suffered from charge traps due to imperfections at or near the surface or gate structures. Traps could be reduced by a black-magic process called UV-flooding, or by uniformly illuminating the CCD at a low level prior to the exposure of the astronomical source (*pre-flashing*). Both processes effectively filled in the traps with some charge.

- Buried Channel CCDs. Since these have largely replaced the original Surface Channel CCDs, the above discussion of CCD structure already silently assumed a buried channel type. The only drawback of the buried channel design is that the storage capacity of each pixel is reduced by a factor 3–4 with respect to a surface channel design.

- Front-illuminated CCDs. In these, photons have to pass through the gates to reach the photosensitive bulk silicon. Since the gates are not entirely transparent, and their transparency depends on wavelength, the peak QE of front-illuminated CCDs is $\sim$70%, and their sensitivity steeply declines below 400 nm. The native thickness of the silicon substrate below the gate structures (typically $\sim$300 μm), does make these devices suitable for detecting far-red and near-infrared light up to $\sim$1.1 μm, but it also makes them more susceptible to charge dumped by the passage of cosmic rays.

- Back-side illuminated CCDs. The bulk silicon substrate of these CCDs is chemically etched away after manufacture. The remaining thickness is a mere $\sim$15 μm. The gate-side (front) of the device is then mounted on a rigid substrate, allowing the device to be illuminated from the back. Since the incident photons no longer have to pass through the gates, peak QEs of $\gtrsim$ 90% are easilly achievable, and the sensitivity to blue and near-UV light is significantly increased, while the susceptibility to cosmic ray induced signal is decreased. Drawbacks are a reduced sensitivity at the red end of optical spectrum, that the etched surface is not as flat resulting in larger variations in sensitivity, and that the remaining layer(s) of material are sufficiently thin to cause interference within the device when illuminated with (near)monochromatic light (e.g., night-sky emission lines).

- Interline and frame transfer CCDs. Light-shielded columns are either interspersed or added to one end of the CCD. During readout, the contents of one column (or the entire illuminated portion of the CCD) are rapidly shifted into the "spare" columns, without the slow step of sampling their stored charge. While the active columns are integrating on the next exposure, the spare columns are being read-out. Only half of the entire array is used to collect photons.

- Anti-blooming CCDs. An additional gate structure is added to the CCD that prevents charge from full-well saturated pixels to migrate to adjacent pixels along a column (called *bleeding*). Instead, any excess charge is drained off and lost. This allows long exposures on faint sources within fields dominated by bright sources (e.g., foreground stars). Drawbacks are, that $\sim$30% of the active pixel area is lost, reducing the peak QE of these devices, and that the total pixel well depth is reduced by a factor of $\sim$2. The tiny gaps in sensitivity between adjacent pixels can also affect the spatial resolution.

- Multipinned phase (MPP) CCDs. The much lower dark current at higher operating temperatures in these devices is achieved by a different electronic design and pixel structures, inversion of all three clock voltages and additional semiconductor doping (with Boron or Phosphorus) of the phase 3 gate. The drawback is a reduced well-depth.

- Othogonal Transfer CCDs (OTCCDs). Charge can be shifted both vertically along columns and horizontally along rows thanks to a spatially complex four-phase design, where the channel stops of regular CCDs become a fourth gate. Two of the (center) gates are triangular in shape, and the remaining two are split into pairs and rectangular in shape, surrounding the center gates (see Fig. 2.7 of Howell 2006). The additional electronic structures result in a lower QE and charge trapping can be more of an issue. By shifting the charges *while exposing*, image motions can be corrected, or PSF's can be actively shaped for high-precision photometry.

- Low Light Level CCDs (L3CCDs). The serial output register is extended and the gate voltages over the extended pixels are much higher (40–60 V), resulting in a finite probability (1–2%) of kicking loose another electrong (avalanche amplification). The total amplification factor scales as $(1+p)^N$, with $p \sim$ 1-2% and N the number of pixels in the extended register (typically ∼500). Since the read-noise of the output amplifier is now leveraged against the amplified signal instead of against the original signal, the S/N is sufficiently high to count individual photo-electrons (=incident photons). Fig. 8 shows this extended register for an E2V L3CCD.

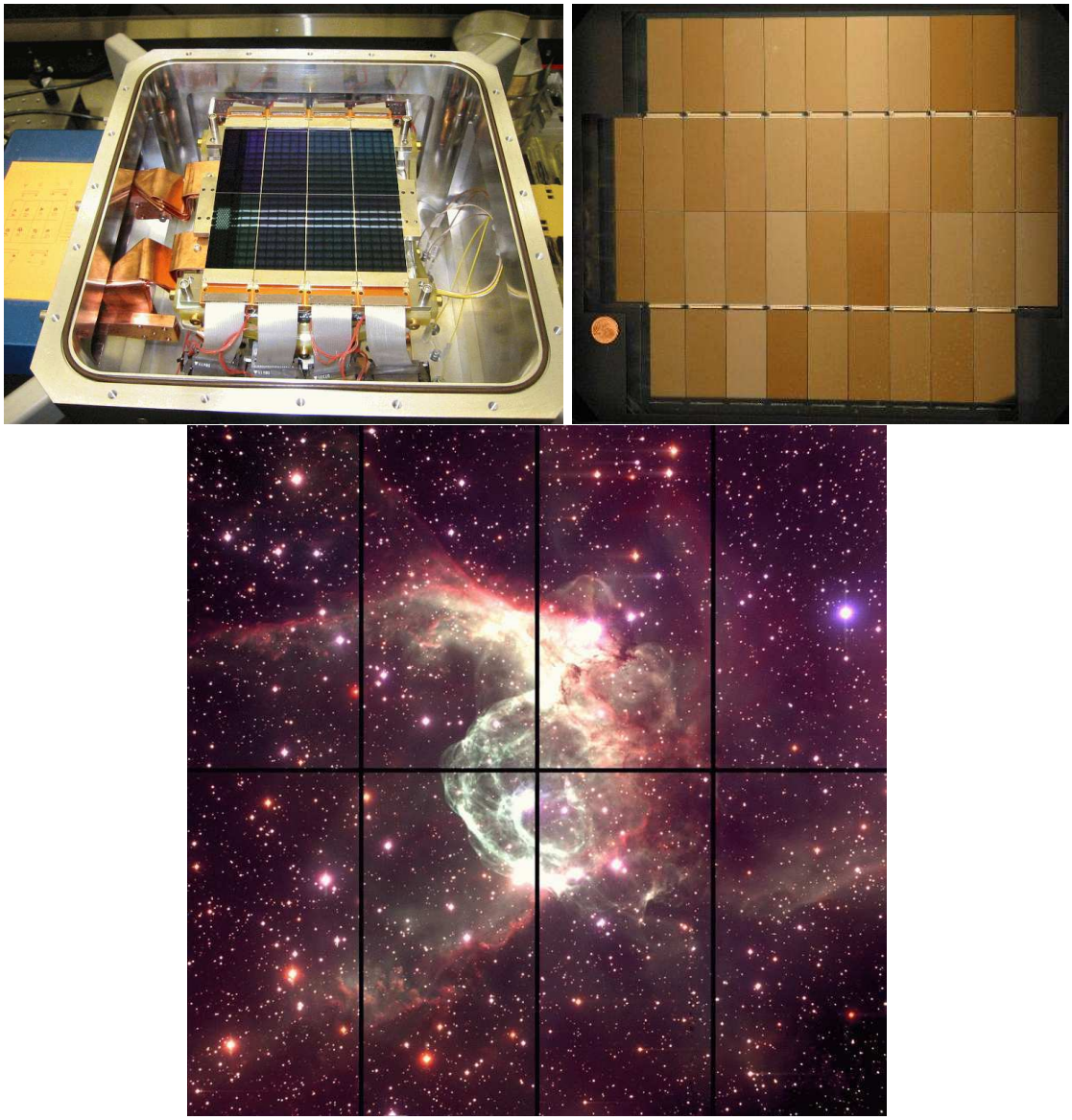


Figure 11: Example of large arrays of CCDs: (a) Magellan/IMACS, an array of 4×2 large-format CCDs. *Courtesy: OCIW*; (b) CFHT/Megacam, an array of a staggering 40! CCDs. *Courtesy: Canada-France-Hawaii Telescope/2003*. These CCD cameras allow imaging over $\sim 0.5^\circ \times 0.5^\circ$ and $\sim 0.94^\circ \times 0.94^\circ$. (c) An example of a wide-field Magellan/IMACS image. Note that the areal coverage is not complete: to fill the gaps ("butt cracks") between the CCDs, several dithered exposures need to be taken. *Courtesy: Danny Steeghs*